

Effects of Confidence Visualization in Automated Assembly Instructions

Kazuma Mori*

Nara Institute of Science and Technology

Isidro III Butaslac

Nara Institute of Science and Technology

Taishi Sawabe

Nara Institute of Science and Technology

Hirokazu Kato

Nara Institute of Science and Technology

Alexander Plopski

Graz University of Technology

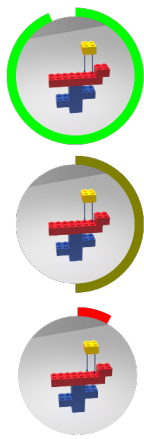


Figure 1: Our automatic assembly instruction system shares the confidence of the current instruction to the user. **Left:** The confidence level indicator surrounding the task instructions changes color continuously from red (low), to yellow (medium), to green (high), representing the three levels of confidence used in the experiment. **Right:** A user wearing a HoloLens 2 assembling LEGO blocks, with the AR panel displaying task instructions positioned in front of them.

ABSTRACT

Automatic Augmented Reality (AR)-based work support systems seamlessly display the next step instruction when the current task step completion is detected, all without requiring explicit user interaction. However, whenever the progress estimation fails it can lead to confusion and loss of trust in the system. To support the user's understanding of the reliability of the displayed instructions, we display the confidence of the current step. To explore the effects the presented confidence has on user interaction with the system, we conducted a Wizard-of-Oz style user study with 12 participants. Participants had to assemble a LEGO model with and without confidence visualization, where the confidence for each step was pre-defined. Our results showed that while the visualization did not increase the user's performance, participants preferred having the confidence displayed to them, especially when it was low. Our findings provide insights into the design of enhancing the user's perception of uncertainty in current step detection in automated instruction systems.

Index Terms: augmented reality, state awareness, work support system, uncertainty visualization

1 INTRODUCTION

Augmented Reality (AR) work support systems provide intuitive visual guidance by overlaying digital information onto the physical environment [1, 2]. And AR enables the integration of virtual information directly into the user's field of view, maintaining

situational awareness without requiring users to shift their attention between the task and a separate screen. This reduces the need for complex hand-eye coordination, decreasing cognitive load and improving work efficiency compared to traditional manuals. However, many existing AR systems require users to manually report the completion of each task step [4]. This not only interrupts the workflow and reduces efficiency but also increases the risk of task errors due to misreporting. To address these issues, automatic recognition of task completion and automatic updating of instructions based on this recognition is necessary.

Advancements in image detection technology using deep learning have made it possible to automatically recognize task status in real-time [3]. However, it is not possible to predict a step with perfect certainty, making erroneous instructions inevitable thus leading to user confusion or task errors.

By conveying the confidence of the recognition in the displayed state, our system can help users make an informed decision about the reliability of the displayed instructions. We hope that by seeing the degree of confidence, users would have an easier time detecting erroneous information. We evaluate how the displayed confidence affects users and their trust in the system through a Wizard-of-Oz style study that controls when and how erroneous instructions are presented.

2 SYSTEM

We developed a HoloLens 2 assembly instruction system that presented LEGO task instructions as an AR panel in front of the users (Figure 1). The AR panel includes task instructions for the current step, an update button, and a movement bar. Users could move the AR panel to any desired position using the movement bar. When they believe that the instruction was incorrect, they could press the update button, prompting the system to display the correct instruction when the incorrect one is displayed.

*e-mail: mori.kazuma.mj2@is.naist.jp

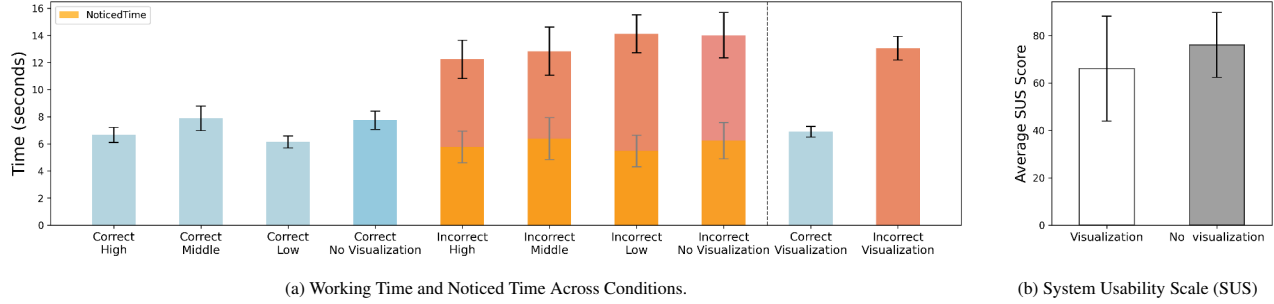


Figure 2: User study results.

The task instructions were surrounded by a confidence indicator. This visualization was preferred in a preliminary study with seven participants. The length of the indicator corresponded to the confidence of the system with no visible bar meaning 0 confidence and 360-degree bar meaning 100% confidence in the displayed step. Furthermore, the color of the indicator changed based on the confidence at the current step, ranging from red (low confidence) to yellow (medium confidence) and green (high confidence).

3 USER STUDY

We designed a within-subjects experiment where participants assembled a 30 step LEGO model, divided into two halves, one with confidence visualization and one without. To simulate state detection, we employed the Wizard-of-Oz method: a researcher observed the user's actions and manually triggered the progression. To ensure a balanced design, the six erroneous instructions were divided into two conditions with and without confidence displayed. In each half, 3 of the 15 steps included predefined erroneous instructions (one each with high, medium, and low confidence in the visualization condition), while the remaining steps followed a balanced random order. This design resulted in 20% incorrect instructions, consistent with prior studies [3].

We compared how participants performed when the confidence was shown and not shown alongside the assembly instructions. To mitigate learning effects, the conditions were alternated. Half of the participants experienced the confidence visualization first, while the other half followed the opposite order. Before continuing with the task for the second half of the assembly, participants were asked to answer the System Usability Scale (SUS) and NASA-TLX questionnaires. At the end of the experiment, participants were also asked about their preferred method, free-form feedback, and demographics information. The order of the conditions and the order of the confidence levels for the incorrect steps were counterbalanced. The experiment took 15 minutes on average. The study was approved by the Ethics Committee of the Nara Institute of Science and Technology (2024-I-39).

We recruited twelve participants (9 male, 3 female, mean age = 24.8 years, $sd = 1.7$) to perform a simple LEGO assembly task from the local university. They rated their familiarity with AR as 1.92 ($sd = 1.44$) on a scale from 1 (Never Used AR) to 5 (Expert Level).

4 RESULTS

We show the time participants took to complete a step in Figure 2a. A Wilcoxon signed-rank test, suitable for non-normally distributed data, was used to analyze execution times under different visualization conditions. The test found no significant differences in execution time for correct ($W = 2436.0, p = 0.120$) or incorrect steps ($W = 36.0, p = 0.850$), nor in the time to detect errors ($W = 38.0, p = 0.970$). For the System Usability Scale (SUS) (Figure 2b), the mean score was 66.04 ($SD = 22.12$) with confidence visualization and 76.04 ($SD = 13.71$) without it. A Wilcoxon signed-

rank test showed no significant differences ($W = 15.0, p = 0.199$). Similarly, for the NASA-TLX total scores, the mean was 38.68 ($SD = 13.11$) with confidence visualization and 34.31 ($SD = 10.87$) without it, also showing no significant differences ($W = 35.0, p = 0.791$).

Regarding user preferences, participants were evenly split between preferring the presence and absence of the confidence level display, with six participants preferring confidence visualization and six preferring its absence. One participant who preferred not displaying confidence commented that high-confidence erroneous instructions increased stress, while low-confidence displays suggested potential errors and reduced stress for two participants. Confidence visualization helped a participant understand the system, but others found decreasing confidence levels led to nervousness.

5 DISCUSSION AND FUTURE WORK

While our results did not reveal statistically significant differences in performance or mental load, this may be due to the simplicity of the task, which reduced participants' reliance on the system. However, in complex tasks, confidence visualization could enhance the user's perception of uncertainty and improve performance.

To address these findings, adaptive confidence visualization could be implemented, emphasizing low-confidence displays while keeping high-confidence indicators less prominent. Additionally, allowing users to customize feedback display methods could further improve usability and satisfaction. In addition, although the AR 2D panel was used for this study, we plan to explore the impact of confidence visualization on cognitive load when situated in the 3D model.

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